CLINICAL · EAR

Superior canal dehiscence

In superior canal dehiscence, loud sounds or pressure make the world move, and the patient may hear their own eyes moving, their footsteps and their pulse. The audiogram can look exactly like otosclerosis, but the stapedial reflexes are normal: check them before any stapes operation.

AT A GLANCE

- Sound and pressure cause vertigo; body sounds are too loud.
- Normal reflexes and a low cVEMP threshold are the clue.
- Oblique CT confirms.
- Plug or resurface if disabling.

1 CHAPTER ONE Overview

A third window into the inner ear.

1.1 Definition

A defect in the bone over the superior semicircular canal, where it rises towards the floor of the middle cranial fossa, leaving the membranous canal covered only by dura.

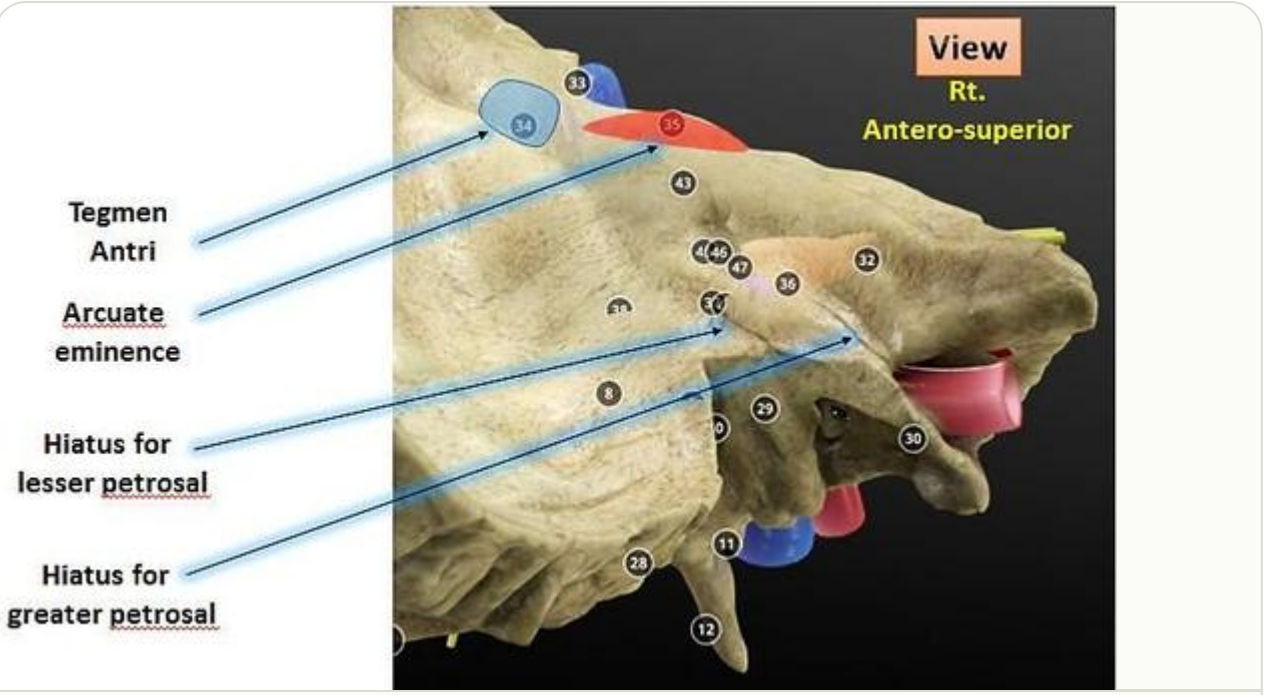


FIGURE 1 The anterior surface of the petrous bone with the arcuate eminence, over the superior canal.

Anatomy atlas · ENT Steps p. 21

1.2 Cause

Probably a developmental thinness of the bone, with the final layer lost later in life through minor trauma, straining or the pulsation of the brain. It is often bilateral on CT, though usually symptomatic on one side.

1.3 The third window

Normally the inner ear has two mobile windows, the oval and the round. The dehiscence adds a third, so sound and pressure energy escape through it. Pressure in the middle ear or the cranial cavity now moves endolymph in the superior canal, causing vertigo; sound energy is diverted away from the cochlea, causing an apparent conductive loss; and bone-conducted sound is heard more loudly.

2 CHAPTER TWO Clinical features

Sound, pressure and the patient's own body.

2.1 Vestibular

- **Tullio phenomenon:** vertigo or oscillopsia triggered by loud sound.
- **Hennebert's sign:** vertigo with pressure in the canal, on Valsalva, coughing or straining.
- Chronic imbalance.

The eye movements induced lie in the plane of the superior canal.

2.2 Auditory

- **Autophony:** the patient's own voice sounds loud.
- **Hyperacusis for bone-conducted sounds:** hearing their footsteps, chewing, heartbeat, or even their eye movements.
- Pulsatile and gaze-evoked tinnitus.
- Aural fullness.

3 CHAPTER THREE Investigations

The clues.

3.1 Audiogram

A low-frequency air-bone gap, with bone conduction thresholds that may be better than 0 dB (supranormal). Stapedial reflexes are *present*, which an otosclerotic or other middle ear conductive loss would abolish.

3.2 Vestibular evoked myogenic potentials

The cervical VEMP threshold is abnormally low and the ocular VEMP amplitude abnormally high. A true middle ear conductive loss normally abolishes VEMPs, so their presence with a gap is a strong clue.

3.3 Electrocochleography

May show a raised SP to AP ratio.

3.4 CT

High-resolution CT reformatted in the plane of the superior canal (Pöschl view) confirms the dehiscence. Thin bone can look dehiscent on CT, so imaging is read with the symptoms and VEMPs.

4 CHAPTER FOUR Differential diagnosis

Mimics.

HOW TO TELL THEM APART

CONDITION	DISTINGUISHING FEATURES
Otosclerosis	Stapedial reflexes absent, VEMPs absent, no vestibular symptoms with sound
Patulous Eustachian tube	Autophony with breathing, drum moves with respiration, no air-bone gap
Perilymph fistula	Follows surgery, trauma or straining; fluctuating hearing
Ménière's disease	Attacks of vertigo for hours, fluctuating low-frequency sensorineural loss
Otosyphilis	Positive serology, Hennebert's sign

5 CHAPTER FIVE Management

Only if it is disabling.

5.1 Conservative

Reassurance and explanation; avoiding triggers such as loud noise and straining. Many patients need nothing more.

5.2 Surgery

For disabling vertigo or autophony: *plugging* the canal or *resurfacing* the defect, through the middle cranial fossa or a transmastoid approach. Plugging is more reliable for vestibular symptoms but sacrifices the canal. Round window reinforcement is an alternative in some centres.

OPERATIVE GUIDES IN THEATRE 1 operation

Superior Semicircular Canal Dehiscence
Superior canal dehiscence

5.3 Risks

Sensorineural hearing loss and vestibular loss after repair, and the usual risks of a middle fossa craniotomy.

BEFORE STAPES SURGERY

A conductive loss with a normal drum and present stapedial reflexes is not otosclerosis. Operating on the stapes in superior canal dehiscence does not help and may harm.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The third window** and how it explains each symptom.
- **Tullio phenomenon** and Hennebert's sign.
- **Conductive gap with present reflexes,** and low VEMP thresholds.
- **CT in the plane of the canal.**
- **Plugging or resurfacing.**

6.2 Common pitfalls

COMMON PITFALLS

- Offering stapes surgery for a conductive loss with present reflexes.
- Diagnosing from CT alone, without symptoms and VEMPs.

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IMAGING

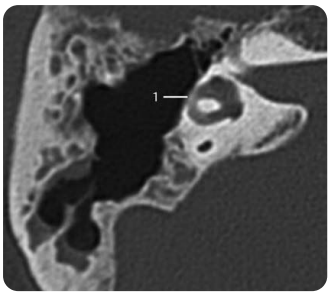
On imaging

The imaging of this topic: key findings, differentials and representative scans.

MIDDLE EAR & MASTOID · AFTER SURGERY

Iatrogenic Fenestration

- In fenestrated patients, no actual compression due to masses will be seen.
- Beware of suction of the cavity for cleaning. This must be performed very carefully to prevent severe vertigo and inner ear dysfunction.



CT, axial. Before the availability of microsurgical techniques and the use of ossicular replacement prostheses, mobilization of the footplate was…